

Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau

TEAM ID	SWTID-2026-6767
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	2 MARKS

3.2 – Solution Requirement

The Electric Vehicle solution is a Tableau-based platform that helps people understand Electric Vehicle charge levels, battery performance, energy consumption and driving range. This system takes raw Electric Vehicle data. Turns it into easy-to-understand visual insights through interactive dashboards and reports. The Electric Vehicle solution must give users a picture of how their vehicle is performing how well it is charging and how the battery is being used.

The system should be able to work with Electric Vehicle datasets that have information like battery percentage how long it takes to charge, charging station details distance travelled, energy consumption, vehicle model and battery health. The Electric Vehicle solution must take this data. Organize it to create accurate visualizations that help users monitor their Electric Vehicle performance in real time or look at it historically.

The platform should have dashboards for things like analysing battery charge predicting range looking at charging behaviour monitoring energy consumption and comparing Electric Vehicle performance. Users should be able to filter the data look closer at points and customize the views to fit their needs. The platform should use charts, graphs, maps and key performance indicators to help users understand the data and make good decisions.

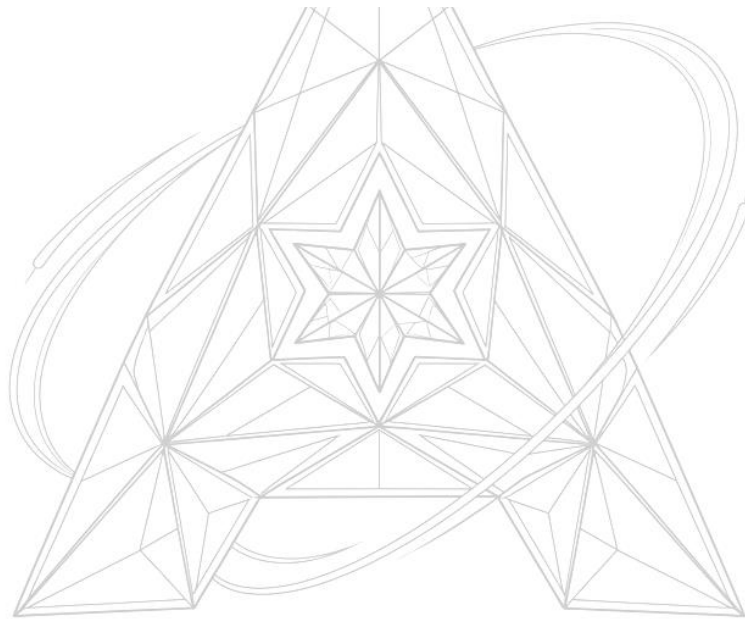
The Electric Vehicle solution should be able to find trends and patterns related to battery discharge how often the Electric Vehicle is charged, charging efficiency and range variations in conditions. It should help users see how things like temperature, driving habits, road conditions and charging methods affect the performance of the Electric Vehicle. The system should also be able to compare Electric Vehicle models to see which ones are the most efficient and effective.

The visualization tool should be easy to use, able to handle a lot of Electric Vehicle data without slowing down and give consistent information in a way that is easy to understand. The dashboards should give users access to important information so they can make informed decisions about how to use and charge their Electric Vehicle.

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The Electric Vehicle solution should help Electric Vehicle owners by reducing their worry, about running out of charge improving battery management and optimizing charging schedules. Manufacturers can use the insights to make Electric Vehicles and improve battery technology while policymakers and infrastructure planners can use the analysis to improve charging station deployment and support sustainable transportation initiatives.

By using Tableaus visualization capabilities, the proposed Electric Vehicle solution aims to deliver a comprehensive, interactive and data-driven platform that improves Electric Vehicle monitoring, performance evaluation and operational efficiency. This Electric Vehicle solution will give stakeholders the information they need to make decisions and promote the use of Electric Vehicles.



ALPHA
— INNOVATORS —

